



# Performances of Machine Learning in Detecting Glaucoma Using Fundus and Retinal Optical Coherence Tomography Images: A Meta-Analysis

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- **PURPOSE:** To evaluate the performance of machine learning (ML) in detecting glaucoma using fundus and retinal optical coherence tomography (OCT) images.
- **DESIGN:** Meta-analysis.
- **METHODS:** PubMed and EMBASE were searched on August 11, 2021. A bivariate random-effects model was used to pool ML's diagnostic sensitivity, specificity, and area under the curve (AUC). Subgroup analyses were performed based on ML classifier categories and dataset types.
- **RESULTS:** One hundred and five studies (3.3%) were retrieved. Seventy-three (69.5%), 30 (28.6%), and 2 (1.9%) studies tested ML using fundus, OCT, and both image types, respectively. Total testing data numbers were 197,174 for fundus and 16,039 for OCT. Overall, ML showed excellent performances for both fundus (pooled sensitivity = 0.92 [95% CI, 0.91-0.93]; specificity = 0.93 [95% CI, 0.91-0.94]; and AUC = 0.97 [95% CI, 0.95-0.98]) and OCT (pooled sensitivity = 0.90 [95% CI, 0.86-0.92]; specificity = 0.91 [95% CI, 0.89-0.92]; and AUC = 0.96 [95% CI, 0.93-0.97]). ML performed similarly using all data and external data for fundus and the external test result of OCT was less robust (AUC = 0.87). When comparing different classifier categories, although support vector machine showed the highest performance (pooled sensitivity, specificity, and AUC ranges, 0.92-0.96, 0.95-0.97, and 0.96-0.99, respectively), results by neural network and others were still good (pooled sensitivity, speci-

ficity, and AUC ranges, 0.88-0.93, 0.90-0.93, 0.95-0.97, respectively). When analyzed based on dataset types, ML demonstrated consistent performances on clinical datasets (fundus AUC = 0.98 [95% CI, 0.97-0.99] and OCT AUC = 0.95 [95% CI, 0.93-0.97]).

- **CONCLUSIONS:** Performance of ML in detecting glaucoma compares favorably to that of experts and is promising for clinical application. Future prospective studies are needed to better evaluate its real-world utility. (Am J Ophthalmol 2022;237: 1–12. © 2021 Elsevier Inc. All rights reserved.)

**G**LAUCOMA, A LEADING CAUSE OF BLINDNESS characterized by an insidious onset, is often first diagnosed only after irreversible vision impairment has occurred. Therefore, the identification of signs of glaucoma through clinical examinations is important.<sup>1</sup> In addition to measurements of intraocular pressure and the visual field, fundus photography and optical coherence tomography (OCT) are 2 other diagnostic modalities used to detect and monitor glaucoma.<sup>1-3</sup>

With an increasing prevalence and irreversible loss of visual function,<sup>4,5</sup> an efficient, sensitive, and specific tool for accurate detection and screening of glaucoma is needed. Because glaucoma evaluation by clinicians can be time- and labor-consuming, skills-demanding, and prone to inconsistency or subjectivity,<sup>6,7</sup> the application of automated image analysis by machine learning (ML) has been investigated as a tool to enhance glaucoma detection.

ML uses existing data to train and build a model that automatically performs certain tasks on a new dataset, providing a less human-dependent approach for medical image interpretation.<sup>8</sup> Ophthalmology employed ML early to classify healthy and diseased images,<sup>9</sup> and excellent results have been achieved for several common ocular diseases.<sup>10</sup> In addition to conventional ML algorithms, such as support vector machine (SVM); random forest; and k-nearest neighbor; deep learning, a newer ML technique that uses the neural network (NN) has transformed automated image analysis.<sup>11-13</sup> One notable example is the superior accuracy of NN in detecting diabetic retinopathy compared with hu-

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man experts,<sup>14</sup> which led to the US Food and Drug Administration's approval of the first automatic diabetic retinopathy screening machine in 2018.<sup>15</sup>

Although many studies have shown positive results of glaucoma image classification by means of ML,<sup>16-18</sup> evidence of its overall performance on this task has been limited. In this study, we examined the diagnostic performance of ML systematically, as applied in glaucoma detection using fundus photographs and retinal OCT images, with additional analyses of the accuracy of different ML classifier categories and its potential real-world applicability.

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## METHODS

This meta-analysis was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. Institutional Review Board approval and informed consent were not required, as all data were extracted from past publications. All research adhered to the Declaration of Helsinki.

• **SEARCH METHODS FOR IDENTIFYING STUDIES:** A literature search for relevant studies published through August 11, 2021 was performed in PubMed and EMBASE. The major search combination terms were keywords associated with glaucoma (eg, *glaucoma*, *open angle glaucoma*, and *glaucomatous optic neuropathy*) crossed with keywords associated with machine learning (eg, *artificial intelligence*, *deep learning*, and *machine learning*). No time or language restriction was applied. The detailed search strategy is provided in the Supplemental Material.

• **ELIGIBILITY CRITERIA AND STUDY SELECTION:** Study selection, data collection, and quality assessment were performed by 2 authors (J.H.W., J.W.L.). After removing duplicates, title and abstracts were screened for eligibility based on the publication or study type and the reported outcomes. Eligibility criteria was full-length journal articles presenting relevant results about the primary outcome, which was the performance of ML in differentiating glaucoma from healthy participants by means of automated image analysis using fundus photographs or retinal OCT images. Case reports, guidelines, conference papers, letters, editorials, and review articles were excluded. Studies that failed to provide results of the primary outcome, including those predicting glaucoma onset, detecting glaucoma progression, or differentiating glaucoma from other ocular diseases, were also excluded. Studies that assessed nonbinary classification outcomes were also considered ineligible. When multiple studies were derived from the same research group with substantial overlap in the testing session, only the most recent study with the best results was included. Detailed information of other exclusion criteria and review protocol are summarized

in the Supplemental Material. Retrieved studies with accessible full articles then underwent full-text review.

• **DATA COLLECTION AND RISK OF BIAS ASSESSMENT:** Data collection was performed during full-text review. The following variables were extracted from each study, if available: first author, published year, country, image type, major ML classifier, total experimental data size (image or eye), sources of training and validation data, total testing data size (image or eye), glaucomatous testing data size (image or eye), ground truth, validation independency or externality, and the sensitivity, specificity, and area under receiver operating characteristic (AUROC) of the best-performing ML algorithm's results derived from each validation or testing dataset. Results derived from different datasets or image types in a study were documented separately and considered independent data. Details of methods for data collection were summarized in the Supplemental Material. All included full-text articles were examined for irrelevant outcomes and data insufficiency, defined as failure to provide sufficient information to construct a  $2 \times 2$  contingency table for the testing results. After data extraction, studies were first categorized into "fundus," "OCT," or "both" based on the image type used, and further classified into 3 ML categories based on the major classifier applied, that is, SVM, NN, and others (all other types of ML technique).

The quality of each included study was assessed using the Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2) tool. QUADAS-2 contains the following 4 domains of assessment: patient selection, index test, reference standard, and flow and timing. The risk of bias for all 4 domains and general applicability for the first 3 domains were examined (Supplemental Material). A study was considered to have a low risk of bias in 1 domain if at least one-half of the extracted data met the requirements of QUADAS-2.

• **DATA SYNTHESIS AND ANALYSIS:** Meta-analysis of the diagnostic performance of ML in detecting glaucoma was performed with a bivariate random-effects model. Because a study might experiment on more than 1 validation or testing datasets, the performance of ML was analyzed using data as the unit (eg, if 3 studies performed glaucoma detection using 5 datasets, a total of 5 data elements will be included). Data of different image types were analyzed separately. Pooled sensitivity, specificity, positive and negative likelihood ratios (LRs), and AUROC were calculated, and results were summarized using hierarchical summary receiver operating characteristic curve plots. The bivariate method modeled the logit-transformed sensitivity and specificity simultaneously to account for the inherent negative correlation arising from the variable thresholds across different studies. Heterogeneity was examined using forest plots and Cochran Q statistics, with  $I^2$  statistic to quantify variation of effect size attributable to between-study heterogeneity. The presence and effect of publication bias were ex-

amined using the Deeks' funnel plots. A trim-and-fill odds ratio was reported to impute the hypothetical missing studies if publication bias presents.

To examine the accuracy of different ML methods, subgroup performance of the 3 ML classifier categories (ie, NN, SVM, and others) was analyzed. To explore whether ML may be helpful in clinical settings, the consistency of its performance on different types of testing datasets was also analyzed. In this subgroup analysis, the validation and testing datasets were classified into the following 3 types: "clinic" (hospital and clinic-based data), "research" (datasets established for longitudinal research purpose), or "public" (public benchmark datasets) based on the purpose and source of data. All analyses were conducted with the "metandi" and "midas" package in Stata, Version 16.0MP (StataCorp). A 2-sided *P* value of .05 indicates statistical significance.

## RESULTS

- **LITERATURE SEARCH:** The literature search yielded 1565 and 1622 records from PubMed and EMBASE, respectively. After excluding 819 duplicates, titles and abstracts of 2368 studies were screened for eligibility, with 274 excluded for ineligible publication or study type, 1912 for irrelevant outcome, 7 for overlapping testing cohort, and 16 for inaccessible full text. One hundred fifty-nine studies went through full-text screening, of which 17 were excluded for irrelevant results, 36 for data insufficiency, and 1 for overlapping testing cohort. Finally, 105 studies (3.3%) were included for data analysis. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses flowchart of study selection is presented in Supplemental Figure 1.

- **STUDY CHARACTERISTICS:** The Supplemental Table summarizes the relevant characteristics of included studies examining ML classifier's performance on detecting glaucoma using automated image analysis. Of all included studies, 73 (69.5%) used fundus photographs,<sup>18-90</sup> 30 (28.6%) used retinal OCT images,<sup>17,91-119</sup> and 2 (1.9%) tested ML using both types of images.<sup>120,121</sup> Fifty-three studies (50.5%) performed at least 1 session of independent validation, and 24 (22.9%) performed at least 1 session of external validation. Most studies (*n* = 55 [52.4%]) used NN as the major ML classifier, 20 (19.0%) used SVM, and 30 (28.6%) used others. The most frequently used classifier methods in studies categorized as "Others" included classification and regression tree, *k*-nearest neighbor, and random forest. The funnel plots for all included data are presented in Supplemental Figure 2, with a trim-and-fill plot provided for fundus data. The forest plots with heterogeneity results are presented in Supplemental Figure 3.

- **QUALITY ASSESSMENT:** Results of quality assessments are summarized in Supplemental Figure 4. Most studies had

low risk of bias and concern of applicability for reference standard, but only approximately one-half were low risk in the patient selection domain. Twelve studies (11%) did not report a clear source of testing data and were graded high risk for patient selection, and 37 studies (35%) had unclear risk due to small data size, potentially biased data composition, and/or using public datasets without available description. For the index test, most studies were graded with unclear risk due to the use of resampling method during ML validation, and 17 (16%) had high risk due to probable data overlap between training and testing sessions. Because many studies used public datasets, of which the flow and timing of dataset construction were unavailable, more than 40% of the studies were graded with unclear risk in this domain. Studies that did not describe the testing data source and validation process were graded as high risk (*n* = 11 [10%]).

- **OVERALL PERFORMANCE OF ML IN DETECTING GLAUCOMA:** Table 1 summarizes the overall performance of ML in detecting glaucoma based on image types, and the summary receiver operating characteristic plots are shown in the Figure. There was a total of 109 data elements for fundus photographs (data size: *n* = 197,174; 37918 [19%] with glaucoma). Using all data, the pooled mean sensitivity, specificity, and AUROC were 0.92 (95% CI, 0.91-0.93), 0.93 (95% CI, 0.91-0.94), and 0.97 (95% CI, 0.95-0.98), respectively. Similar performance was derived using data from independent datasets (data size: *n* = 178,162; 29545 [17%] with glaucoma) and external datasets (data size: *n* = 123,376; 16863 [14%] with glaucoma). The pooled sensitivity, specificity, and AUROC were 0.91 (95% CI, 0.89-0.93), 0.91 (95% CI, 0.89-0.93), and 0.97 (95% CI, 0.95-0.98) for the former, and 0.90 (95% CI, 0.87-0.93), 0.90 (95% CI, 0.86-0.92), and 0.96 (95% CI, 0.93-0.97) for the latter, respectively.

For retinal OCT images, 38 data elements were included (data size: *n* = 16,039; 9468 [59%] with glaucoma). The pooled mean sensitivity, specificity, and AUROC of all OCT data were 0.90 (95% CI, 0.86-0.92), 0.91 (95% CI, 0.89-0.92), and 0.96 (95% CI, 0.93-0.97), respectively. Although the results of independent datasets (data size: *n* = 11,303; 7018 [62%] with glaucoma) were similar to those of all data (sensitivity = 0.90; 95% CI, 0.85-0.93; specificity = 0.90; 95% CI, 0.87-0.92; AUROC = 0.95; 95% CI, 0.92-0.96), a lower sensitivity (0.86; 95% CI, 0.76-0.92), specificity (0.86; 95% CI, 0.84-0.88), and AUROC (0.87; 95% CI, 0.84-0.90) were noticed for results derived using external datasets (data size: *n* = 3385; 2206 [65%] with glaucoma), which might be related to the fewer data available.

- **PERFORMANCE OF ML BASED ON MAJOR CLASSIFIER CATEGORY:** Table 2 summarizes the performance of different categories of ML classifier in detecting glaucoma. Sixty-seven, 18, and 24 fundus data elements used NN, SVM,

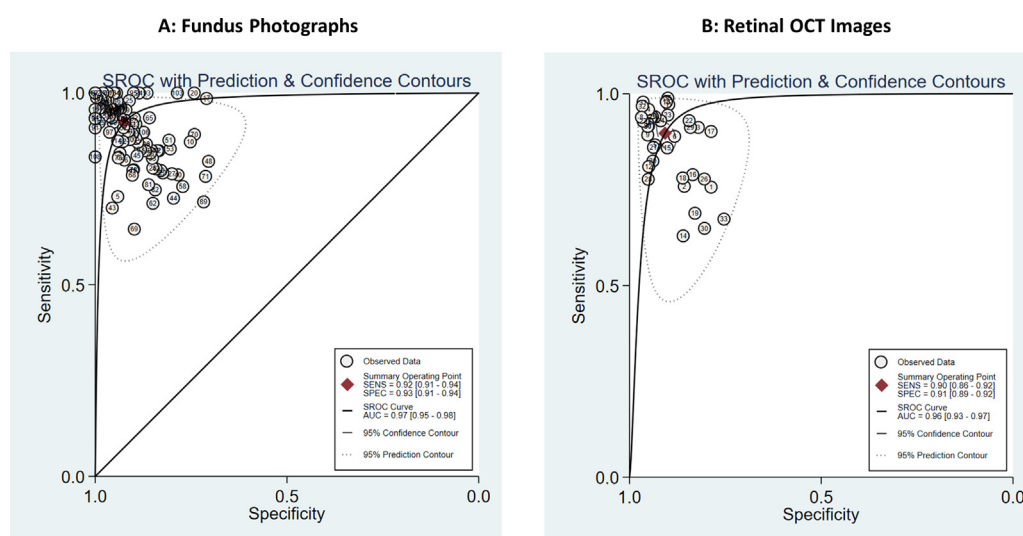
**TABLE 1. Overall Performance of Machine Learning in Detecting Glaucoma Based on Image Types**

| Variable                 | Data, n | Data Size (% With Glaucoma) | Sensitivity (95% CI) | Specificity (95% CI) | LR+ (95% CI)        | LR- (95% CI)        | AUROC (95% CI)      |
|--------------------------|---------|-----------------------------|----------------------|----------------------|---------------------|---------------------|---------------------|
| <b>Fundus</b>            |         |                             |                      |                      |                     |                     |                     |
| All data                 | 109     | 197,174 (19)                | 0.92<br>(0.91-0.93)  | 0.93<br>(0.91-0.94)  | 12.3<br>(10.4-14.7) | 0.08<br>(0.07-0.10) | 0.97<br>(0.95-0.98) |
| Independent <sup>a</sup> | 62      | 178,162 (17)                | 0.91<br>(0.89-0.93)  | 0.91<br>(0.89-0.93)  | 10.5<br>(8.5-13.1)  | 0.09<br>(0.08-0.12) | 0.97<br>(0.95-0.98) |
| External <sup>b</sup>    | 30      | 123,376 (14)                | 0.90<br>(0.87-0.93)  | 0.90<br>(0.86-0.92)  | 8.6 (6.3-11.7)      | 0.11<br>(0.08-0.15) | 0.96<br>(0.93-0.97) |
| <b>OCT</b>               |         |                             |                      |                      |                     |                     |                     |
| All data                 | 38      | 16,039 (59)                 | 0.90<br>(0.86-0.92)  | 0.91<br>(0.89-0.92)  | 9.7 (7.8-12.1)      | 0.11<br>(0.08-0.15) | 0.96<br>(0.93-0.97) |
| Independent              | 18      | 11,303 (62)                 | 0.90<br>(0.85-0.93)  | 0.90<br>(0.87-0.92)  | 8.8 (6.8-11.3)      | 0.12<br>(0.08-0.17) | 0.95<br>(0.92-0.96) |
| External                 | 7       | 3386 (65)                   | 0.86<br>(0.76-0.92)  | 0.86<br>(0.84-0.88)  | 6.2 (5.1-7.5)       | 0.16<br>(0.09-0.29) | 0.87<br>(0.84-0.90) |

AUROC = area under receiver operator characteristic; OCT = optical coherence tomography; LR+ = positive likelihood ratio; LR- = negative likelihood ratio.

<sup>a</sup>Pooled results using data qualifying for validation independency.

<sup>b</sup>Pooled results using data qualifying for validation externality.



**FIGURE 1. Summary receiver operating characteristic (SROC) of all included data using fundus photographs (A) and retinal optical coherence tomography (OCT) images (B). AUC = area under curve; SENS = sensitivity; SPE = specificity.**

and others as the method for major ML classifier, respectively. The best results were derived using SVM (data size:  $n = 3804$ ; 1802 [47%] with glaucoma), with a pooled mean sensitivity of 0.96 (95% CI, 0.93-0.97), specificity of 0.97 (95% CI, 0.94-0.98), and AUROC of 0.99 (0.98, 1.00). NN (data size:  $n = 180,534$ ; 30867 [17%] with glaucoma) and others (data size:  $n = 12,836$ ; 5249 [41%] with glaucoma) had slightly lower but similar performances. The pooled sensitivity, specificity, and AUROC was 0.91 (95% CI, 0.89-0.93), 0.91 (95% CI, 0.89-0.92), and 0.96 (95% CI, 0.94-0.98) for NN; and 0.93 (95% CI, 0.89-0.95), 0.93

(95% CI, 0.91-0.95), and 0.97 (95% CI, 0.9600.98) for others.

For retinal OCT data elements, 20 used NN (data size:  $n = 12,863$ ; 7940 [62%] with glaucoma), 5 used SVM (data size:  $n = 829$ ; 398 [48%] with glaucoma), and 13 used others (data size:  $n = 2347$ ; 1130 [48%] with glaucoma) as the method for major ML classifier. Although the AUROC was similar for all 3 categories (range, 0.95-0.96), SVM still showed a higher pooled sensitivity (0.92; 95% CI, 0.84- 0.96) and specificity (0.95; 95% CI, 0.93-0.97) compared with NN (sensitivity = 0.88; 95% CI, 0.82-0.91;



**TABLE 2.** Performance of Different Categories of Machine Learning Classifier in Detecting Glaucoma

| ML Category   | Data, n | Data Size (% With Glaucoma) | Sensitivity (95% CI) | Specificity (95% CI) | LR+ (95% CI)        | LR- (95% CI)        | AUROC (95% CI)      |
|---------------|---------|-----------------------------|----------------------|----------------------|---------------------|---------------------|---------------------|
| <b>Fundus</b> |         |                             |                      |                      |                     |                     |                     |
| NN            | 67      | 180,534 (17)                | 0.91<br>(0.89-0.93)  | 0.91<br>(0.89-0.92)  | 9.9 (8.1-11.9)      | 0.10<br>(0.08-0.12) | 0.96<br>(0.94-0.98) |
| SVM           | 18      | 3804 (47)                   | 0.96<br>(0.93-0.97)  | 0.97<br>(0.94-0.98)  | 27.7<br>(16.3-47.1) | 0.05<br>(0.03-0.07) | 0.99<br>(0.98-1.00) |
| Others        | 24      | 12,836 (41)                 | 0.93<br>(0.89-0.95)  | 0.93<br>(0.91-0.95)  | 14.1<br>(10.1-19.7) | 0.08<br>(0.05-0.12) | 0.97<br>(0.96-0.98) |
| <b>OCT</b>    |         |                             |                      |                      |                     |                     |                     |
| NN            | 20      | 12,863 (62)                 | 0.88<br>(0.82-0.91)  | 0.90<br>(0.87-0.92)  | 8.6 (6.5-11.3)      | 0.14<br>(0.09-0.20) | 0.95<br>(0.92-0.96) |
| SVM           | 5       | 829 (48)                    | 0.92<br>(0.84-0.96)  | 0.95<br>(0.93-0.97)  | 19.1<br>(12.1-30.0) | 0.08<br>(0.04-0.17) | 0.96<br>(0.93-0.97) |
| Others        | 13      | 2347 (48)                   | 0.92<br>(0.86-0.95)  | 0.90<br>(0.86-0.93)  | 8.9 (6.1-13.1)      | 0.90<br>(0.06-0.16) | 0.96<br>(0.94-0.97) |

AUROC = area under receiver operator characteristic; LR+ = positive likelihood ratio; LR- = negative likelihood ratio; ML = machine learning; NN = neural network; OCT = optical coherence tomography; SVM = support vector machine.

**TABLE 3.** Performance of Machine Learning in Detecting Glaucoma Based on Testing Datasets Types

| Dataset Type  | Data, n | Data Size (% With Glaucoma) | Sensitivity (95% CI) | Specificity (95% CI) | LR- (95% CI)        | LR+ (95% CI)        | AUROC (95% CI)      |
|---------------|---------|-----------------------------|----------------------|----------------------|---------------------|---------------------|---------------------|
| <b>Fundus</b> |         |                             |                      |                      |                     |                     |                     |
| Clinic        | 46      | 128,404 (22)                | 0.94<br>(0.93-0.95)  | 0.94<br>(0.92-0.95)  | 15.6<br>(12.3-19.7) | 0.06<br>(0.05-0.08) | 0.98<br>(0.97-0.99) |
| Research      | 8       | 42,886 (4)                  | 0.84<br>(0.80-0.87)  | 0.86<br>(0.81-0.90)  | 6.1 (4.3-8.6)       | 0.19<br>(0.15-0.24) | 0.91<br>(0.88-0.93) |
| Public        | 50      | 23,751 (32)                 | 0.91<br>(0.88-0.93)  | 0.91<br>(0.88-0.93)  | 9.8 (7.7-12.5)      | 0.10<br>(0.08-0.14) | 0.96<br>(0.94-0.97) |
| <b>OCT</b>    |         |                             |                      |                      |                     |                     |                     |
| Clinic        | 26      | 13,553 (61)                 | 0.90<br>(0.85-0.93)  | 0.90<br>(0.87-0.92)  | 8.9 (7.0-11.4)      | 0.12<br>(0.08-0.17) | 0.95<br>(0.93-0.97) |
| Research      | 7       | 2050 (44)                   | 0.87<br>(0.77-0.93)  | 0.91<br>(0.84-0.95)  | 9.3 (5.2-16.7)      | 0.14<br>(0.07-0.27) | 0.95<br>(0.93-0.97) |

AUROC = area under receiver operator characteristic; LR+ = positive likelihood ratio, LR- = negative likelihood ratio; OCT = optical coherence tomography.

specificity = 0.90; 95% CI, 0.89-0.92) and others (sensitivity = 0.92; 95% CI, 0.86-0.95; specificity = 0.90; 95% CI, 0.86-0.93).

• **PERFORMANCE OF ML BASED ON TESTING DATASET TYPES:** Table 3 summarizes the results of ML based on different testing dataset types. Most studies used clinical (46 data elements, data size: n = 128,404; 27881 [22%] with glaucoma) and public datasets (50 data elements, data size: n = 23,751; 7490 [32%] with glaucoma) for validation, and the performance of ML on clinical data was better than on public benchmark data. The pooled sensitivities for clinical and public datasets were 0.94 (95% CI, 0.93-0.95) and 0.91 (0.88, 0.93), specificities were 0.94 (0.92, 0.95) and 0.91

(0.88, 0.93), and AUROC were 0.98 (95% CI, 0.97-0.99) and 0.96 (95% CI, 0.94-0.97), respectively. In contrast, a lower performance was noticed when longitudinal research data (8 data elements, data size: n = 42,886; 1661 [4%] with glaucoma) were used for testing (sensitivity = 0.84; 95% CI, 0.80-0.87; specificity = 0.86; 95% CI, 0.81-0.90; AUROC = 0.91; 95% CI, 0.88-0.93).

For OCT studies, 26 data elements were from the clinic (data size: n = 13,553; 8319 [61%] with glaucoma) and 7 were from longitudinal research (data size: n = 2050; 908 [44%] with glaucoma). Because there was only 1 OCT data element that used a public dataset, analysis was not performed for this type of dataset. The pooled AUROC and specificity for clinical and research datasets were similar

(AUROC = 0.95; 95% CI, 0.93-0.97 for both; specificities = 0.90; 95% CI, 0.87-0.92 for clinic and 0.91; 95% CI, 0.84-0.95 for research), although the former showed a slightly higher sensitivity (0.90; 95% CI, 0.85-0.93 vs 0.87; 95% CI, 0.77-0.93) of ML.

## DISCUSSION

This study investigated the performances of ML in detecting glaucoma using retinal OCT and fundus photographs. Overall, ML showed an excellent diagnostic performance using both types of images, as evidenced by the high sensitivity, specificity, and AUROC. ML performed similarly using all data and external data when analyzing fundus photographs. In contrast, the external test result of OCT was less robust, possibly due to less availability of data. When comparing different ML classifier methods, SVM showed the highest performance among the 3 categories for both image types. Although the performances by NN and others were slightly inferior, the diagnostic accuracy was still good. In subgroup analysis based on dataset types, ML demonstrated a consistent and robust performance when tested on clinical datasets. Our results suggest that ML may be useful for detection of glaucoma.

In addition to the common issues of fundus imaging process, glaucoma diagnosis using fundus photographs was known to be observer-dependent and inherently variable.<sup>122,123</sup> For optic disc evaluation, a past review revealed an overall sensitivity ranging from 65% to 77% and a specificity ranging from 59% to 98%.<sup>124</sup> Using stereoscopic disc photo, the maximal sensitivity achieved by ophthalmologists was still approximately 75%, with a classification rate of about 80%.<sup>125</sup> In comparison, physicians showed better agreement and accuracy diagnosing glaucoma using OCT parameters, with the sensitivities and specificities mostly ranging from 70% to 95%.<sup>126-129</sup> However, for cases at earlier stages or with lesser visual field impairment, significantly lower sensitivity and AUROC were observed.<sup>129,130</sup> It is notable that most studies evaluating human experts' performances were conducted under ideal conditions with no time limitation, which may not be readily achieved in clinical settings.

As already known, the high efficiency and reliability are some major advantages of ML compared with manual image interpretation. Moreover, based on our results, the accuracy of ML in detecting glaucoma is likely comparable with that of human experts, as shown by the high overall AUROC of >95% for both fundus and OCT images. Although information about human vs ML was sparse for OCT image interpretation,<sup>91</sup> some studies performed prospective, head-to-head comparisons between ML and ophthalmologists using fundus photographs, and the performance of ML was comparable, sometimes even superior, to that of ophthalmologists.<sup>19-23,25,28,56</sup> Of note, in the current study, most of the data for both types of images had a sensitivity of >90%,

which is significantly higher than the average of human observers. This supports the usefulness of ML in clinical examination because a higher sensitivity represents a better chance for patients with disease to be identified, which is particularly beneficial for irreversible disease like glaucoma. Furthermore, some studies provided additional results of ML in differentiating healthy people and early-stage glaucoma, including mild glaucoma, glaucoma suspects, and preperimetric glaucoma. With most AUROC being >85%, especially in OCT studies, it is optimistic that ML may be helpful in early identification of patients in need of medical intervention.<sup>24,92-95</sup>

To investigate whether a particular ML technique may be the state-of-the-art method for glaucoma detection, subgroup analysis was performed based on the major ML classifier categories. To our surprise, NN did not show any better performance than that seen in other tasks,<sup>14,131,132</sup> although its accuracy was still satisfactory. With deep learning being used increasingly for image analysis,<sup>133</sup> its greatest strength, compared with traditional ML, is the multi-layered NN structure; this makes the training process less labor-dependent and more efficient.<sup>11,133</sup> However, because NN develops its own rules for task performing, its performance can be influenced greatly by the quantity and quality of data used at the development stage.<sup>24,30,134-136</sup> Thus, the less ideal training data characteristics of some included NN studies might have contributed to lower pooled results. The varying hidden-layer structures across studies might be another reason. Nevertheless, many large-scale NN studies have shown excellent individual results.<sup>26-29,45,92,94</sup> Therefore, it is likely that a better evaluation of NN's overall performance can be achieved when more high-quality and larger studies become available. Another noteworthy point is that some SVM and others studies applied NN during the feature extraction or early classification stages,<sup>28,31-33,98,120</sup> and the hybrid method might have led to better pooled results that surpass the original accuracy of these conventional methods. This suggests the usefulness of NN not only as the major classifier method, but also as a complementing technique during ML model development.

One important question when evaluating ML is whether its high performance can be generalized to the real world. To address this, we evaluated ML's performance on external datasets and performed subgroup analysis based on dataset types. Although the externally validated results of fundus studies were similar to those using all data, a lower performance was noticed for OCT studies. However, the included OCT data size was significantly smaller than that of fundus, and only 4 OCT studies (7 data elements) performed external validation,<sup>91,92,96,98</sup> 3 of which applied NN as the major classifier. Therefore, it is likely that with more studies and training data, the performance of OCT will improve further. Most importantly, ML showed a high diagnostic performance using clinic-derived fundus and OCT images (sensitivity and specificity >90%, AUROC >95%). The satisfying results derived from clinical data are relevant

to physicians, as they indicate that the performance of ML may be robust in clinical settings, and that this technique may be practically useful. Still, some potential factors affecting ML's performance should be attended, particularly high myopia or pathological myopia, which was reported to be responsible for approximately 50% of false-negative and false-positive results by means of ML.<sup>26,27</sup> Although ML has shown noninferior accuracy compared with ophthalmologists, its role still should be considered complementary to the usual clinical evaluation, and physicians should be more attentive to patients with other ocular conditions that may confound the optic disc characteristics.

Another interesting observation is the relatively lower accuracy of ML when tested on longitudinal research datasets. It is known that, with a 2-gate study design or dataset composition of only healthy versus definite disease cases, an overestimation of a diagnostic test's accuracy can occur.<sup>137</sup> This is especially concerning in studies using cross-sectional public datasets, as the selected images may not be fully reflective of real-world scenario, where cases with mild or nontypical changes may be present and disease prevalence is usually lower.<sup>138</sup> Therefore, although ML showed robust performance on other dataset types, the possibility of an inflated accuracy due to selection bias caused by particular study design and dataset construction cannot be excluded. Still, compared with other dataset types, significantly fewer studies were available for analysis of ML's performance on longitudinal research data, which can lead to a lower estimated accuracy. Considering these points, it is important to further examine ML's performance when more longitudinal studies are available. In addition, because 1 of the study objectives was to evaluate the ML method that will most likely be applied clinically, only the best-performing algorithms and data were included. Although it was expected that the results obtained may be slightly overestimated, this possible source of accuracy inflation should still be minded.

Several limitations of the current study should be noted. First, glaucoma is an umbrella term with different disease subtypes, and it is unknown whether our results can be applied to a particular type or stage of glaucoma. However, with most included studies aiming at detection of any

referable glaucoma, our results are useful for the current stage of ML's application and general image reading purpose. Second, this study had a higher between-study heterogeneity, which might have resulted from the varying image settings, data characteristics, thresholds effect, and ML details. Nevertheless, with the large number of studies included and the heterogeneous nature of ML design, a high heterogeneity when conducting overall evaluation was expected. With only a small portion of data showing performance different from the others in the forest plots, the value of our results should not be undermined. Moreover, results of AUROC and sub-analyses were provided to help address the threshold effect and potential sources of heterogeneity. Third, a lack of transparency with the ML methods was noted in some studies. Although this should not affect the current analysis, clarification of the detailed technique may be clinically beneficial, as knowing which structural features led to better diagnostic performance may provide insight about glaucoma and its management. Last, prospective studies directly examining ML's performance and comparing it with ophthalmologists are still lacking. It should also be noted that the utility of a diagnostic tool may vary based on the scenario of decision making, with consideration of the testing purpose and disease prevalence. Because community- or population-based studies were limited, we cannot determine ML's performance in glaucoma screening for the general public. To have a better evaluation of ML's practical usefulness in both glaucoma detection and screening, more large prospective studies with external validation and well-defined clinical questions are needed.

In conclusion, the performance of current ML technique in detecting glaucoma using fundus and OCT images compares favorably to that of human experts, and its excellent accuracy when tested on clinic-based datasets is promising for clinical application. Nonetheless, it is important to keep in mind the potential risk of inflated accuracy, and the real-world utility of ML can only be concluded when more large-scale, prospective studies are available. This study provides important evidence for determining the current state of ML for this task, and may serve as the basis for future analyses on its usefulness in glaucoma screening.

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**Data availability:** All data pertinent to this study are provided in the main text or Supplemental Material. No additional data were generated.

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